Day-12

Memories.

FIFO (First in First Out)

- Data is read in the same order it is written.
- Used as a temporary data storage buffer.
- Helps transfer data between modules operating at different speeds.

FIFO Input & Output Ports.

Inputs.

Signal	Function
Clk	Clock signal
rst	Active high ~~speed~~ reset
wr_en	Write Enable
rd_en	Read Enable
din	Input Data.

Outputs.

Signal	Function.
dout	Output Data
full	Indicates FIFO is full
empty	Indicates FIFO is empty.

FIFO Working.

Write Operation.

Write occurs when.

- wr_en=1
- FIFO is NOT full

During writing:

- Data is stored into FIFO memory
- Write Pointer increments..

Read Operation
Read occurs when:
- $rd\_en = 1$
- FIFO is NOT empty.

During reading:
- Data is read from FIFO
- Read Pointer increments.

FIFO Depth calulation.
FIFO Depth = Total Data written − Data Read During writing Time.

1) FA > FB (No Idle cycle)
- Writing frequency = 80MHz
- Reading frequency = 50MHz
- Burst length = 120

Calulation.
- Writing Time = 120×12.5ns = 1500ns
- Reading Time per Data = 20ns
- Data Read = 1500/20 = 75
- FIFO Depth = 120−75 = 45

FIFO Depth = 45

2) FA < FB
- Write frequency = 30MHz
- Read frequency = 50MHz.

Since reading is faster than writing
FIFO Depth = 1.

3) FA = FB
- Write frequency = 50MHz
- Read frequency = 50MHz
- Write Data = 1
- Read Idle = 3

Calculation:

- Total Write Time = 1800N
- Data Read = 60
- FIFO Depth = 120-60 = 60.

4) FA = FB

- If clocks are perfectly synchronized.
  - FIFO is not required.
- If there is a phase difference
  - FIFO depth 1 is sufficient